Year 13 Maths - Functions

Similar questions - Composite and inverse functions

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The function f is defined by $\mathrm{f}(x) = x^2 - 6x + 12$ for $x \geqslant k$.

(a) Write $\mathrm{f}(x)$ in the form $(x-p)^2 + q$ and state the least value of k for which f^{-1} exists. **[3]**

Take k to be this least value in the rest of the question.

(b) Find $\mathrm{f}^{-1}(x)$, stating its domain and range. **[3]**

(c) Find $\mathrm{ff}(5)$. **[2]**

(d) Solve $\mathrm{f}(x) = x$ and explain why your solutions are also solutions of $\mathrm{f}(x) = \mathrm{f}^{-1}(x)$. **[4]**

2 $\mathrm{f}(x) = 2x + 1$ and $\mathrm{g}(x) = x^2 - 3$, both for all real x.

(a) Find expressions for $\mathrm{fg}(x)$ and $\mathrm{gf}(x)$. **[2]**

(b) Solve $\mathrm{gf}(x) - \mathrm{fg}(x) = 9$. **[3]**

(c) State the range of fg. **[1]**

(d) Explain why g^{-1} does not exist, and describe how the domain of g could be restricted so that it does. **[2]**

Question 1

(a) $f(x) = (x-3)^2 + 3$. The vertex is at $x = 3$, so the least k is 3

(b) $y = (x-3)^2 + 3$, so $x = 3 + \sqrt{y-3}$ (positive root as $x \geqslant 3$). $f^{-1}(x) = 3 + \sqrt{x-3}$, domain $x \geqslant 3$, range $f^{-1}(x) \geqslant 3$

(c) $f(5) = 7$, $f(7) = 19$, so $ff(5) = 19$

(d) $x^2 - 7x + 12 = 0$, $(x-3)(x-4) = 0$, so $x = 3$ or $x = 4$. If $f(x) = x$ then applying f^{-1} to both sides gives $x = f^{-1}(x)$, so $f(x) = x = f^{-1}(x)$ (the graphs of f and f^{-1} are reflections in $y = x$ and meet on it)

Question 2

(a) $fg(x) = 2(x^2 - 3) + 1 = 2x^2 - 5$. $gf(x) = (2x+1)^2 - 3 = 4x^2 + 4x - 2$

(b) $4x^2 + 4x - 2 - 2x^2 + 5 = 9$, so $2x^2 + 4x - 6 = 0$, $x^2 + 2x - 3 = 0$ and $x = 1$ or $x = -3$

(c) $fg(x) = 2x^2 - 5 \geqslant -5$

(d) g is many-one (e.g. $g(2) = g(-2)$), so it has no inverse. Restrict to $x \geqslant 0$ (or $x \leqslant 0$) so it is one-one